

# elegant-polimi-thesis documentation

This chapter is the result of the documentation generated from the source code. Since it's not strictly needed to use the template – most of the functions are straightforward – it's quite barebones.

This document contains them all, even those that are not exposed to the user, which are marked with “\_” before.

## All functions

|                                    |    |                                   |    |
|------------------------------------|----|-----------------------------------|----|
| Theses templates .....             | 1  | _spacer .....                     | 12 |
| polimi-thesis .....                | 1  | _stroke-no-border .....           | 12 |
| polimi-article-format-thesis ..... | 3  | _show-field .....                 | 13 |
| polimi-executive-summary .....     | 5  | _show-coadvisor .....             | 13 |
| _blank-toc .....                   | 6  | _poli-header .....                | 13 |
| frontmatter .....                  | 7  | _poli-bg-split .....              | 14 |
| acknowledgements .....             | 7  | _divider-body .....               | 14 |
| mainmatter .....                   | 7  | _divider-bg-args .....            | 14 |
| appendix .....                     | 7  | _basic-subslide-preamble .....    | 14 |
| backmatter .....                   | 7  | _sizes .....                      | 15 |
| _lists .....                       | 7  | _document-state .....             | 15 |
| _lists-entries-style .....         | 8  | _document-type .....              | 15 |
| nomenclature .....                 | 8  | _localization .....               | 15 |
| theorems-init .....                | 9  | bluepoli .....                    | 15 |
| subfigure .....                    | 9  | _pane .....                       | 15 |
| toc .....                          | 9  | _custom-numbering .....           | 15 |
| list-of-figures .....              | 9  | Figures styling .....             | 15 |
| list-of-tables .....               | 10 | _subfigure-caption .....          | 15 |
| theorem .....                      | 10 | _style-figures .....              | 16 |
| proposition .....                  | 10 | Presentation .....                | 17 |
| lemma .....                        | 10 | poli-slide .....                  | 17 |
| remark .....                       | 10 | title-slide .....                 | 17 |
| proof .....                        | 10 | make-outline .....                | 18 |
| Utilities .....                    | 10 | exact-split-slide .....           | 18 |
| _is-page-empty .....               | 10 | split-slide .....                 | 19 |
| _empty-page .....                  | 10 | plot-slide .....                  | 19 |
| _nameref .....                     | 11 | image-slide .....                 | 19 |
| _raggiera-image .....              | 11 | background-slide .....            | 20 |
| banner .....                       | 11 | focus-slide .....                 | 21 |
| tab-numbering .....                | 11 | _new-section-slide .....          | 21 |
| chapter-numbering .....            | 12 | _new-subsection-slide .....       | 21 |
| header-number .....                | 12 | polimi-digital-presentation ..... | 21 |

## Theses templates

### polimi-thesis

The main thesis formatting function.

## Parameters

```
polimi-thesis(  
  title: str,  
  author: str,  
  advisor: str,  
  coadvisor: str arr,  
  academic-year: str,  
  tutor: str,  
  cycle: str,  
  chair: str,  
  student-id: str,  
  course: str,  
  frontispiece: "phd" "deib-phd" "cs-eng-master" "classical-master",  
  custom-logo: image,  
  body  
) -> content
```

**title** str

Title of the thesis.

Default: "Thesis Title"

**author** str

Author of the thesis.

Default: none

**advisor** str

Advisor of the thesis.

Default: none

**coadvisor** str or arr

Coadvisor(s) of the thesis.

Default: none

**academic-year** str

Academic year of the thesis. If empty, defaults to “2025-2026”.

Default: ""

**tutor** str

Tutor of the thesis.

Default: none

**cycle** `str`

Cycle of the thesis.

Default: `none`

**chair** `str`

Chair of the thesis.

Default: `none`

**student-id** `str`

Student ID.

Default: `none`

**course** `str`

Student course.

Default: `none`

**frontispiece** `"phd" or "deib-phd" or "cs-eng-master" or "classical-master"`

Frontispiece of the thesis. Can be either: phd, deib-phd, cs-eng-master or classical-master.

Default: `"phd"`

**custom-logo** `image`

Custom logo of the thesis.

Default: `none`

## **polimi-article-format-thesis**

The thesis article format styling function.

## Parameters

```
polimi-article-format-thesis(  
  title: str,  
  author: str,  
  advisor: str,  
  coadvisor: str array,  
  academic-year: str,  
  student-id: str,  
  course: str,  
  abstract: content,  
  keywords: str,  
  logo: path,  
  body  
) -> content
```

**title** str

Title of the thesis.

Default: "Thesis Title"

**author** str

Author of the thesis.

Default: none

**advisor** str

Advisor of the thesis.

Default: none

**coadvisor** str or array

Coadvisor(s) of the thesis.

Default: none

**academic-year** str

Academic year of the thesis. If empty, defaults to "2025-2026".

Default: ""

**student-id** str

Student ID.

Default: none

**course** `str`

Student course.

Default: `none`

**abstract** `content`

Abstract.

Default: `[]`

**keywords** `str`

Keywords, that appear below the abstract (and in the PDF metadata).

Default: `none`

**logo** `path`

Logo of the thesis.

Default: `image("img/logo_ingegneria.svg", width: 83mm)`

## **polimi-executive-summary**

The executive summary styling function.

### **Parameters**

```
polimi-executive-summary(  
  title: str,  
  author: str,  
  advisor: str,  
  coadvisor: str array,  
  academic-year: str,  
  course: str,  
  logo: path,  
  body  
) -> content
```

**title** `str`

Title of the thesis.

Default: `"Thesis Title"`

**author** `str`

Author of the thesis.

Default: `none`

**advisor** `str`

Advisor of the thesis.

Default: `none`

**coadvisor** `str` or `array`

Coadvisor(s) of the thesis.

Default: `none`

**academic-year** `str`

Academic year of the thesis. If empty, defaults to “2025-2026”.

Default: `""`

**course** `str`

Student course.

Default: `none`

**logo** `path`

Logo of the thesis.

Default: `image("img/logo_ingegneria.svg", width: 83mm)`

## **\_blank-toc**

Helper function to manually add blank space above an outline entry (similar to  $\text{\LaTeX}$ 's `\addtocontents{toc}{\vspace{1em}}`).

### **Parameters**

`_blank-toc(space: length) -> content`

**space** `length`

Vertical space to add before the next outline entry.

Default: `1em`

## frontmatter

Frontmatter section. Similar to  $\text{\LaTeX}$ 's `\frontmatter`, it is meant to be only used in the thesis. It sets the page numbering to "i" and numbering: `none` for headings.

### Parameters

`frontmatter`(body) -> `content`

## acknowledgements

Acknowledgements section. It sets numbering: `none` for headings.

### Parameters

`acknowledgements`(body) -> `content`

## mainmatter

Mainmatter section. Similar to  $\text{\LaTeX}$ 's `\mainmatter`, it is meant to be only used in the thesis. It sets to page numbering to "1", heading numbering to `"1.1" + h()` and resets the page counter.

### Parameters

`mainmatter`(body) -> `content`

## appendix

Appendix section. Similar to  $\text{\LaTeX}$ 's `\appendix`. It sets heading numbering to `"A.1"` and resets their counter.

### Parameters

`appendix`(body) -> `content`

## backmatter

Backmatter section. Similar to  $\text{\LaTeX}$ 's `\backmatter`, it is meant to be only used in the thesis. It sets heading numbering to `none`.

### Parameters

`backmatter`(body) -> `content`

## \_lists

Lists figure to make the list of tables, list of figures to appear in the table of contents.

### Parameters

`_lists`() -> `content`

## **`_lists-entries-style`**

Internal helper function to create the custom lists of figures and table.

### **Parameters**

```
_lists-entries-style(  
  outline-entry: outline-entry ,  
  kind: function  
) -> content
```

**outline-entry** `outline-entry`

Outline entry to edit.

**kind** `function`

The kind of the outline entry element (image or table)

## **nomenclature**

Displays a simple nomenclature with keys and values.

```
#nomenclature((  
  "Polimi": "Politecnico di Milano",  
  "CdL": "Corso di Laurea",  
  "CCS": "Consigli di Corsi di Studio",  
  "CFU": "Crediti Formativi  
Universitari",  
))
```

### **Nomenclature**

|               |                                |
|---------------|--------------------------------|
| <i>POLIMI</i> | Politecnico di Milano          |
| <i>CDL</i>    | Corso di Laurea                |
| <i>CCS</i>    | Consigli di Corsi di Studio    |
| <i>CFU</i>    | Crediti Formativi Universitari |

### **Parameters**

```
nomenclature(  
  dict: dictionary ,  
  indented: bool  
) -> content
```

**dict** `dictionary`

Dictionary that hold keys and values.

**indented** `bool`

Whether to indent or not the nomenclature.

Default: `true`



## theorems-init

Utility function to initialize the theorem environments<sup>1</sup>:

- theorem
- proposition
- lemma
- remark

### Parameters

`theorems-init`(body) -> `content`

## subfigure

Creates a new subfigure with the given arguments and an optional label. From hallon package.

### Parameters

```
subfigure(  
  ..args,  
  outlined: bool,  
  label: str,  
  body  
) -> content
```

**outlined**    `bool`

Whether to outline this subfigure or not.

Default: `false`

**label**    `str`

Unique label to reference this subfigure

Default: `none`

**toc**    `content`

Custom-built `outline()`.

**list-of-figures**    `content`

List of figures. Similar to  $\text{\LaTeX}$ 's `\listoffigures`.

---

<sup>1</sup>Provided by great-theorems package.

## **list-of-tables** content

List of tables. Similar to  $\text{\LaTeX}$ 's `\listoftables`.

## **theorem** content

Theorem block.

## **proposition** content

Proposition block.

## **lemma** content

Lemma block.

## **remark** content

Remark block.

## **proof** content

Proof block.

## Utilities

### **`_is-page-empty`**

Check if a page is empty.

#### Parameters

`_is-page-empty()` -> bool

### **`_empty-page`**

Adds an empty page between an odd page and the next. Used to check when to remove the header and place a raggiera in the bottom left corner.

#### Parameters

`_empty-page()` -> content

## **`_nameref`**

It displays a reference using section name (instead of numbering). From Hallon 0.1.3

### **Parameters**

```
_nameref(label) -> content
```

## **`_raggiera-image`**

Inserts a raggiera, given a specified width.

### **Parameters**

```
_raggiera-image(width: length) -> content
```

**width**    `length`

Width of the raggiera.

## **`banner`**

Draw a banner.

### **Parameters**

```
banner(body: content) -> content
```

**body**    `content`

Body of the banner.

## **`tab-numbering`**

Alternative numbering: "1.1." + h-space.

### **Parameters**

```
tab-numbering(  
  h-space: length,  
  ..n: arguments  
) -> content
```

**h-space**    `length`

Horizontal space.

Default: `0.7em`

```
..n arguments
```

Numbers.

## **chapter-numbering**

Chapter numbering.

### **Parameters**

```
chapter-numbering(..n: arguments) -> content
```

```
..n arguments
```

Numbers.

## **header-number**

Numbering used in the theses header.

### **Parameters**

```
header-number(..n: arguments) -> content
```

```
..n arguments
```

Numbers.

## **\_spacer**

Empty block.

### **Parameters**

```
_spacer(  
  width,  
  height  
) -> content
```

## **\_stroke-no-border**

Internal stroke.

### **Parameters**

```
_stroke-no-border(color) -> stroke
```

## **`_show-field`**

Helper function to detect whether a field is present and, if true, show it.

### **Parameters**

```
_show-field(  
    prefix: str,  
    field: variable,  
    separator: func content  
) -> content
```

**prefix**    `str`

Prefix (e.g. "Prof: ").

**field**    `variable`

Exact field to check (e.g. title).

**separator**    `func` or `content`

Separator between fields.

Default: `linebreak()`

## **`_show-coadvisor`**

Helper function to handle coadvisor(s).

### **Parameters**

```
_show-coadvisor(coadvisor: str arr) -> content
```

**coadvisor**    `str` or `arr`

Coadvisor(s) of the thesis.

## **`_poli-header`**

Custom made header.

### **Parameters**

```
_poli-header(  
    text-args: dict,  
    self  
) -> content
```

**text-args** dict

The text arguments for the title.

Default: (size: 15pt, weight: "regular", fill: rgb("#5e5e5e"))

## **`_poli-bg-split`**

Apply a fill colour to the panes.

### **Parameters**

```
_poli-bg-split() -> content
```

## **`_divider-body`**

New section body, that is title, lettering and chapter number.

### **Parameters**

```
_divider-body(
  title,
  lettering,
  number,
  text-fill
) -> content
```

## **`_divider-bg-args`**

Style the new section background.

### **Parameters**

```
_divider-bg-args(
  fill,
  stroke
) -> content
```

## **`_basic-subslide-preamble`**

Standard subslide preamble.

### **Parameters**

```
_basic-subslide-preamble(
  self,
  inset,
  text-args
) -> content
```

**\_sizes** dict

TeX sizes to match original templates (<https://tex.stackexchange.com/questions/24599/what-point-pt-font-size-are-large-etc>)

**\_document-state** state

The current state (title page, front-,main-,backmatter...)

**\_document-type** state

The current type of document.

**\_localization** dict

Localization dictionary. It could be deprecated since theses are written exclusively in English.

**bluepoli** color

Signature PoliMi colour (■), used in headings and labels.

**\_pane** dict

The background panes used in some scenarios.

**\_custom-numbering** numbering

(1..20) -> (01,02,03,...,20)

## Figures styling

**\_subfigure-caption**

It displays the caption of subfigures. Use “(a)” that will be colored based on colored-headings parameter.

## Parameters

```
_subfigure-caption(  
    colored-caption: bool,  
    parent: figure,  
    it  
) -> content
```

**colored-caption**    `bool`

Whether to apply bluepoli on captions

Default: `true`

**parent**    `figure`

Parent figure of this subfigure

Default: `none`

## \_style-figures

Style-figures handles (optional heading-dependent) numbering of figures and subfigures. Can modify the caption of all figures.

## Parameters

```
_style-figures(  
    colored-caption: bool,  
    heading-levels: int,  
    body  
) -> none
```

**colored-caption**    `bool`

Whether to apply bluepoli on captions ([Figure 1](#)).

Default: `true`

**heading-levels**    `int`

Heading level to account for in the numbering.

Default: `0`



# Presentation

## poli-slide

Basic slide.

### Parameters

```
poli-slide(  
    config: dictionary,  
    repeat: auto bool,  
    setting,  
    composer,  
    ..args,  
    body  
) -> content
```

**config** dictionary

Touying overrides for this slide.

Default: ( : )

**repeat** auto or bool

Whether to repeat this slide.

Default: auto

## title-slide

The title slide. The style can be specified via the corresponding argument. For the background and logos style you must also specify a background and logos.

### Parameters

```
title-slide(  
    style: "default" "white" "logos" "background",  
    background: image,  
    logos: array,  
    config: dictionary  
) -> content
```

**style** "default" or "white" or "logos" or "background"

Style of the title slide.

Default: "default"

**background** image

Required background for the “background” style; otherwise ignored.

Default: none

**logos**    `array`

Required logos for the “logos” style; otherwise ignored.

Default: ()

**config**    `dictionary`

Touying overrides for this slide.

Default: ( : )

## make-outline

Generate the outline based on headings.

### Parameters

```
make-outline(config: dictionary) -> content
```

**config**    `dictionary`

Touying overrides for this slide.

Default: ( : )

## exact-split-slide

Slide exactly split in 2 or 4. Requires content for each split.

### Parameters

```
exact-split-slide(  
  splits: arr,  
  config: dictionary  
) -> content
```

**splits**    `arr`

Content of each split. Usually a title with a small description.

Default: ()

**config**    `dictionary`

Touying overrides for this slide.

Default: ( : )

## split-slide

Split a slide in two panes – similar to header spacing.

### Parameters

```
split-slide(  
  left: content,  
  right: content,  
  config: dictionary  
) -> content
```

**left**    `content`

Left split content.

Default: []

**right**    `content`

Right split content.

Default: []

**config**    `dictionary`

Touying overrides for this slide.

Default: ( : )

## plot-slide

Slide with coloured background. Used for plots.

### Parameters

```
plot-slide(  
  config: dictionary,  
  body  
) -> content
```

**config**    `dictionary`

Touying overrides for this slide.

Default: ( : )

## image-slide

Slide with image on the left, right or as whole background. Deprecated. Use `background-slide()`.

## Parameters

```
image-slide(  
  image: image ,  
  style: "background" "left" ,  
  config: dictionary ,  
  body  
) -> content
```

**image**    image

Image for the slide.

Default: **none**

**style**    "background" or "left"

Style of the image, that is where it will be placed and sized.

Default: "background"

**config**    dictionary

Touying overrides for this slide.

Default: ( : )

## background-slide

Slide with image as whole background.

### Parameters

```
background-slide(  
  image: image ,  
  config: dictionary  
) -> content
```

**image**    image

Image for the slide.

Default: **none**

**config**    dictionary

Touying overrides for this slide.

Default: ( : )

## focus-slide

Last slide. Usually it contains “Thanks for listening!” or similar phrases.

### Parameters

```
focus-slide(  
  lettering: content,  
  config,  
  body  
) -> content
```

**lettering**    content

Bottom lettering.

Default: none

## \_new-section-slide

The so-called “Divisorio”, a new section. Invoked with a level 1 heading.

### Parameters

```
_new-section-slide(  
  config,  
  ..args  
) -> content
```

## \_new-subsection-slide

The other so-called “Divisorio”, but for lvl.2 heading and in a light background. Unused.

### Parameters

```
_new-subsection-slide(  
  config,  
  ..args  
) -> content
```

## polimi-digital-presentation

Main presentation function.

### Parameters

```
polimi-digital-presentation(  
  divider-style: "dark" "light",  
  subtitle-every-style: bool,  
  ..args,  
  body  
) -> content
```

**divider-style** "dark" or "light"

The style of the divider.

Default: "dark"

**subtitle-every-style** bool

Whether to use the subtitle on all title styles.

Default: false